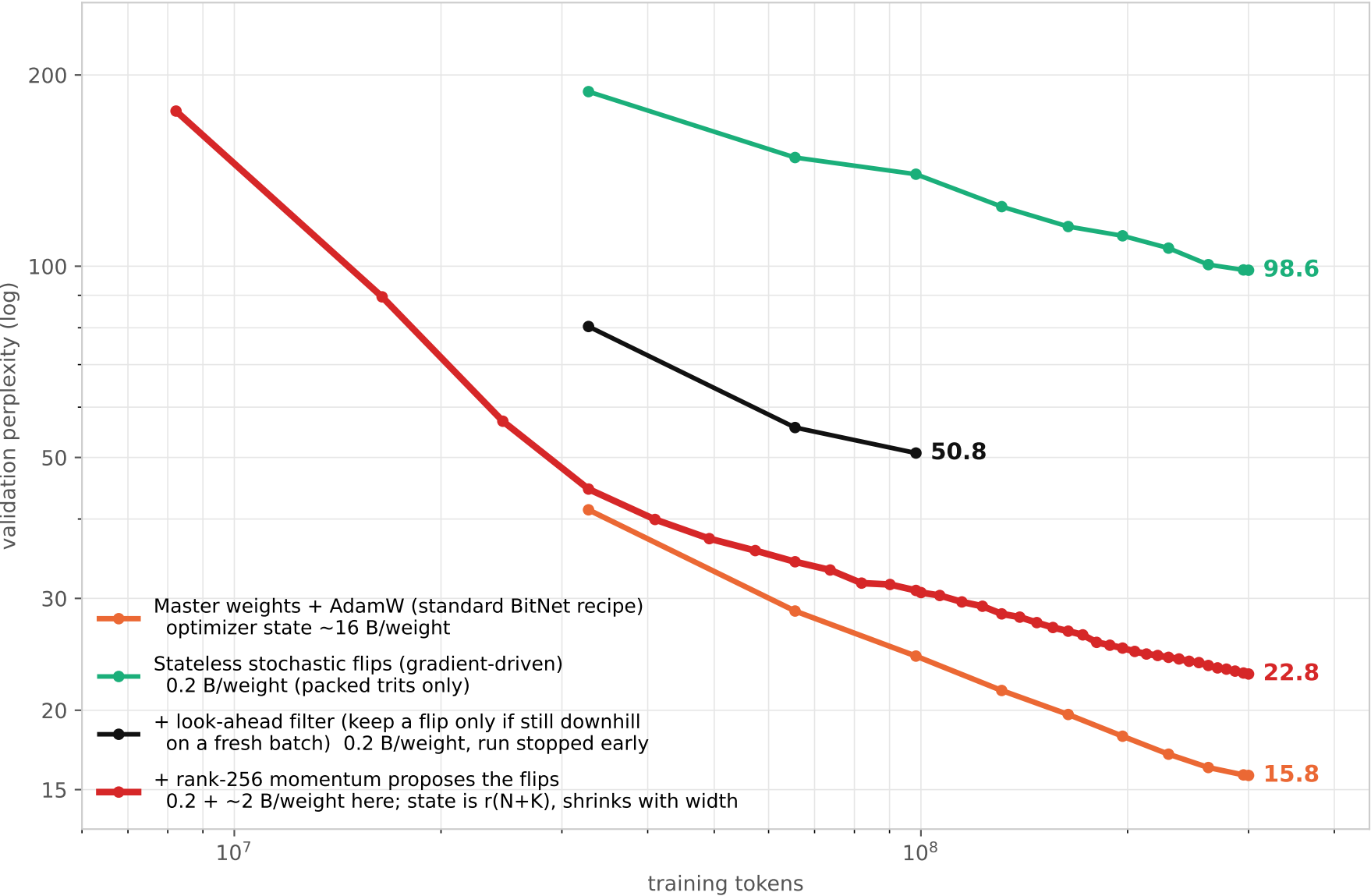


Training ternary {-1, 0, +1} LLMs without full-precision master weights

(a) Validation perplexity, 110M-parameter ternary transformer, WikiText (Llama 32k tokens)



(b) What the momentum does: switch it off at 11M tokens

